CLIENT INVESTMENT & BROKERAGE MANAGEMENT PLATFORM

Cloud-Native Solution Proposal

AWS Cloud • React • API-led • Event-driven Architecture

Document Information	Details
Prepared for	Seaport Global LLP
Prepared by	Essarsystems Pvt Ltd.
Document version	3.0
Date	23 October 2026
Status	Proposal / For Discussion

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1. Executive Summary

This proposal presents a secure, scalable and cloud-native Client Investment & Brokerage Management Platform for managing the end-to-end lifecycle of client investments into financial products and services. The platform is intended to provide a centralised operating capability for investment intake, eligibility and underwriting, pool/allocation management, fund administration, ongoing servicing and reporting.

The proposed solution uses React JS for the web experience and AWS managed services for application hosting, APIs, asynchronous processing, data persistence, security, monitoring and analytics. The architecture is designed around modular business capabilities and event-driven integration so that individual services can scale independently.

The target outcome is a modern platform that reduces manual processing, improves operational control and auditability, enables faster product and service onboarding, and provides a foundation for future digital investment capabilities.

Plan and investment at a glance. Delivery is planned over 26 weeks in six phases with four formal approval gates. The indicative investment is 580 to 850 hours, or $58,000 to $102,000 (see Section 18). Pricing will be refined after discovery and scope confirmation.

2. Business Objectives

- Digitise and streamline the client investment journey from application through ongoing servicing.
- Provide a single platform for managing investment products, applications, transactions and client service activities.
- Automate validation, eligibility, underwriting and approval workflows where appropriate.
- Improve visibility of investment pools, funds, allocations, balances and transactions.
- Provide consistent operational, management, client and regulatory reporting.
- Reduce manual processing, operational risk and duplicated data entry.
- Provide secure API-based integration with internal and third-party systems.
- Create a scalable AWS foundation capable of supporting increased transaction volumes and users.

3. Proposed Solution

The proposed platform is organised into six primary business capabilities. Each capability can be implemented as an independently deployable service or bounded module, while sharing common security, identity, data, integration, monitoring and reporting capabilities.

Module	Primary Responsibility
Ingest	Capture investment requests, client data, documents and inbound information; validate and route requests.
Underwrite	Assess product eligibility, risk, suitability and approval requirements; manage decision workflows.
Pool	Manage investment pools, capacity, allocation and participation rules.
Fund	Administer investments/funds, contributions, allocations, withdrawals, balances, transactions and reconciliation.
Service	Manage ongoing client and investment servicing, amendments, requests, maturity and communications.
Report	Provide client, operational, financial, management, audit and regulatory reporting.

3.1 Target Business Lifecycle

Illustrative lifecycle:

SP User → Ingest → Underwrite → Pool / Allocation → Fund → Service → Report

Reporting will operate as a cross-cutting capability and may consume data from all business modules rather than representing only the final stage of the lifecycle.

3.2 Scope Definition and Scope Decisions

The platform covers the client investment lifecycle for investments into financial products, including credit and asset-backed pools, as detailed per module in Section 4 and Section 18.2. Both sections describe the same scope.

Scope decision required – brokerage execution. The platform name refers to brokerage, but the modules above do not include trade execution. The following capabilities are excluded from the estimate unless Seaport Global confirms them as in scope:

- Order management and routing to execution venues
- Trade capture, confirmation and settlement
- Commissions, fees and rebate calculation
- Broker / introducer onboarding and management
- Market-data feeds for pricing and best-execution evidence

3.3 Success Measures

Targets will be baselined against current operations during discovery and agreed at Gate 1. Proposed measures:

- Application-to-decision cycle time
- Share of applications processed without manual re-keying
- Reconciliation breaks per period and time to resolve
- Time to produce client statements and regulatory reports
- Platform availability and incident volumes against the agreed service levels
- Time to onboard a new product or pool

4. Functional Scope

This section lists capabilities at business level. Items marked as new align the module scope with the detailed capability set in Section 18.2.

Ingest Module

- Client/application onboarding and capture
- Investment request and product selection
- Document upload and metadata management
- API and file-based data ingestion
- Validation, duplicate detection and enrichment

- Application status and workflow tracking
- KYC/AML information capture and integration
- Automated PDF/Excel ingestion with OCR text extraction
- Secure staging with error flagging and exception queues

Underwrite Module

- Product and investment eligibility checks
- Risk and suitability assessment
- Rules-based decisioning
- Approval and manual review workflows
- Exception management
- Decision history and audit trail
- Configurable credit scoring and financial ratio engine, including DSCR
- Underwriter override workflows and term sheet / credit memo generation

Pool Module

- Pool creation and lifecycle management
- Investment/product allocation
- Capacity and participation management
- Allocation rules and controls
- Pool reconciliation and status management
- Asset grouping by yield, maturity, geography and risk rating
- Concentration limit monitoring; multi-tranche structuring
- Pool-level cash flow modelling and stress testing (advanced capability, Phase 4)

Fund Module

- Fund/product administration
- Investment transactions and contributions
- Allocations, withdrawals and redemptions
- Balances and transaction history
- Reconciliation and operational controls
- Valuation and reference information
- Investor ledger, commitment tracking, capital calls and pro-rata allocation
- Disbursement file generation (ACH / Fedwire / SWIFT) and liquidity dashboards

Service Module

- Client and investment account servicing
- Amendments and service requests
- Transfers and withdrawals
- Maturity and lifecycle events

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- Notifications and case management
- Client communication support
- Double-entry sub-ledger for principal, interest, fees and escrows
- Amortisation schedules, payment matching, delinquency tracking and late fees

Report Module

- Client statements and investment reports
- Portfolio and fund reports
- Transaction and operational reports
- Management dashboards
- Regulatory and audit reporting
- Data export and downstream reporting feeds
- Investor portal for statements and tax forms
- Ad-hoc report builder with Excel/PDF export

Cross-cutting Platform Capabilities

- Identity, role-based access and administration console
- Workflow and approval engine, notifications and document management
- Immutable audit trail, configuration and reference-data management

5. Solution Architecture

The target architecture follows cloud-native principles: managed AWS services, API-led integration, event-driven processing, least-privilege security, horizontal scalability and observable services.

5.1 Logical Architecture

The layered view below replaces the text diagram in v3.0. A formal AWS architecture diagram will be produced in the Architecture phase.

Layer	AWS services / components
Users	Seaport Global users, clients and brokers/investors (SP User)
Edge and presentation	Route 53, Amazon CloudFront, AWS WAF and Shield; React / TypeScript web application hosted on Amazon S3
API layer	Amazon API Gateway with Cognito / enterprise IdP authorisation
Business services	Ingest, Underwrite, Pool, Fund and Service services; AWS Step Functions for approval and long-running workflows
Messaging and events	Amazon SQS, Amazon EventBridge (with schema registry), Amazon SNS, Amazon SES

Layer	AWS services / components
Data	Aurora PostgreSQL (Multi-AZ) with RDS Proxy; Amazon S3 with Object Lock for records and documents
Reporting and analytics	Aurora read replica for operational reports; S3, Glue and Athena for analytics; Amazon QuickSight for dashboards
External systems	KYC/AML, credit bureaus, banks and payment rails, reference data – via secure APIs or SFTP
Cross-cutting	IAM, KMS, Secrets Manager, CloudTrail, CloudWatch and X-Ray, GuardDuty, Security Hub, Config, AWS Backup

5.2 AWS Architecture Principles

- Use managed AWS services where practical to minimise infrastructure administration.
- Separate presentation, API, business logic, integration and data layers.
- Use asynchronous messaging for long-running or decoupled processing.
- Design production workloads for multi-AZ resilience.
- Apply least privilege and defence-in-depth security.
- Centralise monitoring, logging, audit and alerting.
- Automate infrastructure and deployment through Infrastructure as Code and CI/CD.

5.3 AWS Foundation and Landing Zone

- AWS Organizations and AWS Control Tower with service control policies (SCPs) as guardrails.
- Separate AWS accounts for security/log archive, shared services and networking, development/test, UAT and production. Production is isolated from all other environments.
- IAM Identity Center for workforce access; no long-lived access keys.
- Centralised, immutable logging (CloudTrail, Config, VPC flow logs) in a dedicated log-archive account.
- Mandatory tagging, budgets and cost allocation reporting from day 1.
- AWS Region(s) chosen to satisfy data-residency requirements (Appendix B). If Seaport Global already operates an AWS organisation, platform accounts will be created within it.

5.4 Resilience and Disaster Recovery

Area	Proposed approach
Compute and network	Multi-AZ deployment of services, load balancing and NAT; no single-AZ dependencies in production.
Database	Aurora PostgreSQL multi-AZ with automated backups and point-in-time recovery; cross-Region backup copies through AWS Backup.
Documents	S3 versioning and cross-Region replication for regulated records.

Area	Proposed approach
DR pattern	To meet the proposed RTO of 4 hours, a pilot-light pattern in a secondary Region is the working assumption. Backup-and-restore only is a lower-cost alternative with a longer RTO. The choice and its AWS cost impact are decided in the Architecture phase.
DR testing	A failover exercise before go-live and at least annually thereafter.

5.5 Design Patterns for Financial Integrity

- Idempotent APIs and message consumers, so retries cannot duplicate transactions.
- Transactional outbox pattern, so database changes and published events cannot diverge.
- Saga / compensation logic through Step Functions for multi-step processes such as capital calls and disbursements.
- Append-only double-entry ledger with reversing entries rather than updates or deletes, plus automated reconciliation.
- Versioned event schemas through the EventBridge schema registry.
- Immutable audit records using S3 Object Lock where record-retention rules require it.

6. Technology Stack

Layer	Proposed Technology	Purpose
Frontend	ReactJS	Responsive web application and client/broker user journeys
CDN / Hosting	Amazon CloudFront + Amazon S3	Secure, scalable static application delivery
API	Amazon API Gateway	Managed API exposure, throttling, authentication integration
Application	Java / Spring Boot on Amazon ECS Fargate for core transactional services; AWS Lambda for event handlers, document processing and integration adapters	Business services and APIs
Workflow	AWS Step Functions	Approvals, sagas and long-running processes
Document intelligence	Amazon Textract	OCR and data extraction from PDFs and scans
Messaging	Amazon SQS	Reliable asynchronous processing
Events	Amazon EventBridge	Domain/integration events and routing
Client communications	Amazon SES	Client emails and notifications
Database	Amazon Aurora PostgreSQL with RDS Proxy	Core transactional data
Documents	Amazon S3	Secure object/document storage
Identity	Amazon Cognito and/or Enterprise IdP	Authentication and federation
Secrets	AWS Secrets Manager	Credentials and secret management

Encryption	AWS KMS (customer-managed keys)	Encryption key management
Monitoring	Amazon CloudWatch, AWS X-Ray / OpenTelemetry	Metrics, logs, alarms and operational dashboards
Audit	AWS CloudTrail	AWS activity and audit logging
Protection	AWS WAF, AWS Shield	Web application protection
Threat detection and compliance	GuardDuty, Security Hub, AWS Config, Amazon Macie, Amazon Inspector	Threat detection, posture management, PII discovery, vulnerability scanning
Backup	AWS Backup	Centralised, cross-Region backup policy
Analytics	Athena, Glue, Amazon QuickSight (Redshift if volumes warrant)	Reporting store and dashboards
Landing zone	AWS Organizations, Control Tower, IAM Identity Center	Multi-account governance
Notifications	Amazon SNS	Operational and support notifications
CI/CD	AWS CodePipeline/CodeBuild or GitHub Actions	Automated build, test and deployment
IaC	AWS CDK / Terraform	Repeatable infrastructure provisioning

7. Security & Compliance

Security will be embedded throughout the architecture and delivery lifecycle. Detailed regulatory requirements will be confirmed during discovery and translated into technical controls.

- Strong authentication and MFA for privileged and appropriate user journeys.
- Role-based access control and least-privilege IAM.
- Encryption in transit using TLS and encryption at rest using AWS KMS.
- Secrets managed through AWS Secrets Manager rather than application configuration.
- Private subnets for sensitive backend resources where appropriate.
- Security Groups, Network ACLs and VPC endpoints based on connectivity requirements.
- AWS WAF for web/API protection.
- CloudTrail and application audit trails for traceability.
- Data retention, archival and deletion policies.
- Environment separation across development, test, UAT and production.
- Regular vulnerability, dependency and security testing.
- Backup, recovery and disaster-recovery controls aligned to agreed RPO/RTO.
- Customer-managed KMS keys with rotation, and field-level protection for sensitive client data.
- Continuous threat detection and posture management (GuardDuty, Security Hub, Config) and PII discovery (Macie).
- Immutable audit and record retention using S3 Object Lock where required.
- Privacy by design: data minimisation, data classification, and a Data Processing Agreement.

7.1 Candidate Regulatory and Assurance Frameworks

The target jurisdiction has not yet been confirmed. The frameworks below are candidates only. Seaport Global compliance and legal stakeholders will confirm which apply.

Area	Candidate requirements (to be confirmed)
Data protection	UK GDPR / EU GDPR; US state privacy laws and GLBA, as applicable
Conduct and records	FCA and MiFID II record-keeping (UK/EU); SEC/FINRA books-and-records rules including SEC Rule 17a-4 (US), where applicable
Financial crime	UK Money Laundering Regulations 2017 / US BSA-AML; sanctions screening
Operational resilience	FCA/PRA operational resilience (UK); DORA (EU), as applicable
Assurance	SOC 2 Type II and ISO 27001 aligned controls; platform evidence to support SOC 1 reporting; AWS Well-Architected Framework and Financial Services Industry Lens
Payments	NACHA rules (ACH), Fedwire and SWIFT message standards; PCI DSS only if card data is handled

8. Data & Integration Architecture

Aurora PostgreSQL is proposed for core transactional data. Amazon S3 will be used for documents, large objects, reports and suitable archival data. Reporting workloads will be isolated from core transaction processing: an Aurora read replica serves operational reports, and S3 with Glue and Athena serves analytical reporting, moving to Amazon Redshift only if volumes or complexity warrant it

8.1 Core Data Domains

- Client
- Investment Account
- Product
- Investment
- Pool
- Fund
- Transaction
- Service Request
- Document
- Report
- Investor / Commitment
- Asset / Borrower
- Tranche
- Capital Call
- Ledger Entry

8.2 Integration Patterns

- REST APIs for synchronous business interactions.
- SQS for reliable asynchronous processing and workload buffering.
- EventBridge for event publication and routing.
- Secure file exchange for agreed batch integrations.
- Third-party APIs for KYC/AML, payment, product and reference-data services as required.
- All integrations are idempotent, versioned and monitored, with dead-letter queues and replay for failed messages.

9. Non-Functional Requirements

Items previously "to be agreed" now carry a proposed target for Seaport Global to confirm or amend in discovery

Area	Proposed Approach / Target
Availability	Proposed: 99.9% monthly production availability excluding agreed maintenance windows; multi-AZ resilience.
Performance	Proposed: API response p95 below 500 ms for interactive requests at agreed peak load; long-running work processed asynchronously. Performance baselines established before UAT.
Scalability	Horizontal scaling using managed/serverless services; capacity based on agreed TPS and concurrent-user volumes.
Security	Least privilege, encryption, MFA, WAF, audit and continuous security controls.
RPO	Proposed: 15 minutes or less for transactional data; confirmed during DR design.
RTO	Proposed: 4 hours or less for critical services; confirmed against the operating model.
Observability	Central metrics, logs, traces where applicable, dashboards and alerts.
Auditability	End-to-end transaction and administrative audit trail.
Maintainability	Modular services, automated deployment, coding standards and documentation.
Accessibility	WCAG 2.1 AA for client-facing and internal web interfaces.
Browser support	Current and previous major versions of Chrome, Edge, Safari and Firefox.
Data residency and retention	Per Appendix B; confirmed in discovery.

10. DevSecOps & Engineering

The engineering model will use automated quality and security gates across the delivery pipeline.

- Source control and branching strategy.
- Automated build and unit tests.
- Static code analysis and dependency vulnerability scanning.
- API and integration test automation.

- Infrastructure as Code.
- Automated environment deployment.
- Release approvals and production change controls.
- Application and infrastructure monitoring.
- One AWS account per environment, with identical IaC promoted from Dev to Test, UAT and Production.
- Pipeline quality gates: unit-test coverage threshold on new code, no critical or high vulnerabilities, and IaC policy checks before deployment.

11. Testing Strategy

- Unit Testing
- Component and API Testing
- Integration / System Integration Testing (SIT)
- End-to-End Testing
- Regression Testing
- User Acceptance Testing (UAT)
- Performance and Load Testing
- Security Testing
- Disaster Recovery Testing
- Production Smoke Testing
- Data migration reconciliation and cutover rehearsal
- Accessibility testing against WCAG 2.1 AA
- Independent penetration test before go-live

Performance testing will be based on agreed transaction volumes, peak TPS, concurrent users and response-time requirements. The final test strategy will be confirmed once workload characteristics and integration dependencies are known.

12. Delivery Approach & Indicative Timeline

Essarsystems recommends a phased, Agile-based delivery model structured around 2-week sprint cycles. This approach ensures continuous stakeholder visibility, early risk mitigation, and iterative delivery of business value

12.1 Delivery Model Principles

- **Agile & Iterative Delivery:** Bi-weekly sprint reviews and product demos to ensure alignment with business requirements.
- **DevSecOps First:** Automated CI/CD pipelines, Infrastructure as Code (AWS CDK / Terraform), and continuous security scanning from Day 1.
- **Modular Progression:** Incremental deployment of business modules (Ingest, Underwrite, Pool, Fund, Service, and Report).

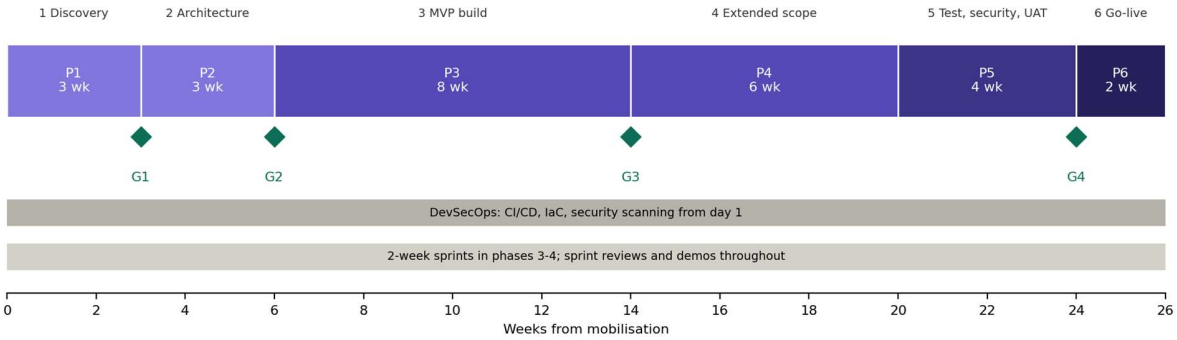
- **Phase Gates:** Formal sign-offs at completion of Discovery, Architecture, MVP, and UAT prior to production deployment.

12.2 Indicative Project Timeline

The roadmap is a single sequential plan of 26 weeks (about six months) in six phases.

Phase	Duration	Key Deliverables & Activities	Exit gate
1. Discovery & Inception	3 weeks	Stakeholder workshops, detailed functional scope, NFR baselining, integration mapping (KYC/AML, market feeds), and prioritised product backlog	Gate 1: scope and backlog
2. Architecture & Solution Design	3 weeks	High-Level Design (HLD), Low-Level Design (LLD), AWS Cloud Architecture and landing zone, API specs (OpenAPI/Swagger), UI/UX wireframes, and data models	Gate 2: design
3. MVP Development	8 weeks	Thin end-to-end slice: Ingest, Underwrite, Pool (core allocation), Fund (core ledger and transactions), basic reports; React Web UI, AWS services, database schemas and primary integrations	Gate 3: MVP acceptance
4. Extended Capabilities	6 weeks	Service module (sub-ledger, amortisation, payments, delinquency), advanced Pool (tranching, cash flow modelling), disbursement formats, investor portal, reporting engine and dashboards, secondary integrations	Continuous (no gate)
5. Testing, Security, Migration & UAT	4 weeks	System Integration Testing (SIT), End-to-End testing, performance/load testing, independent penetration testing, data migration rehearsal, DR test, and User Acceptance Testing (UAT)	Gate 4: UAT acceptance
6. Production Go-Live & Hypercare	2 weeks	Cutover, production deployment, operational readiness, monitoring dashboard setup, administrative handover, and post-launch hypercare support	Go-live and hypercare exit

12.3 Phase Roadmap Summary



If a gate is not approved, work returns to the preceding phase for rework before the next phase starts

The timeline is indicative and will be refined following discovery, product prioritisation, integration assessment, data migration requirements and agreed MVP scope.

12.4 MVP Definition

The MVP is a thin end-to-end slice of the lifecycle in Section 3.1: capture and validate an application (Ingest), decide it (Underwrite), allocate it to a pool (Pool), record the investment and transactions (Fund), and report on it. Pool precedes Fund because the lifecycle requires allocation before funding. The Service module, advanced Pool capabilities, the investor portal and regulatory exports follow in Phase 4.

13. Team Structure & Governance

Role	Primary Responsibility
Product Owner / Business Lead	Prioritisation, business decisions and acceptance
Delivery / Project Manager	Planning, governance, RAID management and status reporting
Business Analyst	Requirements, process mapping and user stories
Solution Architect	End-to-end solution and architecture governance
Technical Architect / Lead Engineer	Detailed technical design and engineering standards
UX / UI Designer	Wireframes, design system and accessibility
React Engineers	Frontend development
Backend Engineers	APIs, business services and integrations
AWS / Cloud Engineer	AWS platform, networking and cloud services
DevOps Engineer	CI/CD, IaC and release automation
QA / Test Engineers	Test automation, SIT, regression and UAT support
Security Specialist	Security architecture and testing
Data / Reporting Engineer	Data pipelines, reporting and analytics

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13.1 Governance and Engagement Model

- Steering committee (Seaport Global sponsor, Essarsystems delivery director) meets monthly and at each gate.
- Weekly status report covering progress, risks, issues, dependencies and decisions (RAID log).
- Bi-weekly sprint reviews and demos with the Product Owner and business stakeholders.
- Formal change control: scope changes are estimated, approved in writing and tracked.
- Escalation path: Delivery Manager, then Delivery Director, then executive sponsors.

13.2 Client Responsibilities

- A named Product Owner with decision authority, available for sprint ceremonies and gate reviews.
- Timely access to business SMEs, compliance and legal stakeholders, and product rules.
- AWS account and access arrangements, network connectivity and enterprise security standards.
- Third-party API specifications, sandbox access and credentials (KYC/AML, banks, data providers).
- UAT participants and prompt defect triage during testing.
- Working-hours overlap: Essarsystems proposes at least four hours of daily overlap with the Seaport Global (EST) working day.

14. Key Deliverables

- Business and functional requirements
- Product backlog, epics, stories and acceptance criteria
- Solution Architecture Document
- High-Level Design (HLD)
- Low-Level Design (LLD)
- AWS architecture and network diagrams
- Data model and database design
- API specifications / Swagger
- Security architecture
- Integration design and sequence diagrams
- React UI/UX implementation
- Backend services and APIs
- Infrastructure as Code
- CI/CD pipelines
- Test strategy and test evidence
- Performance test results
- Operational runbooks and monitoring dashboards
- Production deployment and handover documentation
- AWS landing zone and account structure
- Threat model and independent penetration test report
- Data migration plan, reconciliation report and cutover plan
- DR plan and DR test results
- AWS Well-Architected review report
- Training materials and RACI

15. Assumptions & Dependencies

- Business stakeholders will provide timely access to requirements, product rules and decision criteria.
- External system/API specifications will be available before integration development.
- KYC/AML, payment and other third-party providers will be confirmed during discovery.
- Client will provide required product, fund, pricing and reference-data definitions.
- Data migration requirements will be assessed separately if legacy data must be migrated.
- Regulatory and compliance obligations will be confirmed by the client's compliance/legal stakeholders.
- AWS account structure, network connectivity and enterprise security standards will be agreed before production implementation.
- Commercial estimates will be refined once scope, volumes, integrations and non-functional requirements are baselined.
- Data migration: the estimate covers migration planning and one rehearsal cycle for a single agreed legacy source. Additional sources or complex data cleansing follow change control.
- The target jurisdiction(s) and applicable regulators will be confirmed in discovery.
- Brokerage execution capabilities (Section 3.2) are excluded unless confirmed as in scope.
- The client provides the Product Owner and SMEs described in Section 13.2.
- The estimate assumes one production AWS Region plus a DR Region; further Regions are a change.

16. Key Risks & Mitigations

Risk	Potential Impact	Mitigation
Unclear product/business rules	Rework and delays	Early discovery workshops, rule catalogue and business sign-off
Third-party integration dependency	Schedule impact	Early API validation, mocks and integration milestones
High/unknown transaction volumes	Performance issues	Define TPS/concurrency targets and conduct performance testing
Data quality / migration complexity	Operational issues	Data profiling, validation, reconciliation and phased migration
Security/compliance changes	Design rework	Security and compliance review from architecture phase
Reporting load on transactional DB	Performance degradation	Separate reporting/analytics workload where required
Operational readiness gaps	Production incidents	Runbooks, dashboards, alerting, support training and hypercare
Scope ambiguity (brokerage vs credit/securitisation)	Estimate and design change	Scope decisions in Section 3.2 confirmed at Gate 1
Estimate uncertainty before design is complete	Cost variance	Estimate ranges are indicative and refined after discovery; price confirmed at Gate 2
Jurisdiction and regulator not yet confirmed	Design rework	Confirm in discovery; compliance review at Gate 2
Limited time-zone overlap	Slower decisions	Agreed overlap hours, a named Product Owner and written decision log

17. Support & Operations

To ensure the ongoing reliability, security, high availability, and optimal performance of the **Client Investment & Brokerage Management Platform**, Essarsystems offers a structured, multi-tiered Support and Operations framework. Our model is designed to support a cloud-native, event-driven AWS architecture, minimizing operational friction while safeguarding business continuity.

17.1 Support Tiers & Operating Model

Support operations are structured into progressive tiers to ensure swift issue resolution, minimal disruption, and expert technical escalation when required:

Tier 1: Service Desk & Monitoring (L1)

- **Scope:** Initial point of contact, alert triage, user access administration, routine password/identity resets, and basic workflow query resolution.
- **Actions:** Automated event logging via Amazon CloudWatch alarms, filtering of false positives, and ticketing queue management.

Tier 2: Application & Technical Support (L2)

- **Scope:** Investigation of functional errors, API failures, database query performance anomalies, transaction processing bottlenecks, and integration exception handling.
- **Actions:** Log analysis via CloudWatch/CloudTrail, targeted data patching (subject to audit controls), configuration adjustments, and user guidance.

Tier 3: Engineering & Architecture Escalation (L3)

- **Scope:** Complex defects, code-level bugs, core architecture enhancements, major infrastructure adjustments (IaC/Terraform updates), and security incident response.
- **Actions:** Direct code fixes by React / Spring Boot / Lambda engineers, database tuning on Amazon Aurora PostgreSQL, and AWS infrastructure re-configuration.

17.2 Support Coverage Windows

Coverage Plan	Window	Description	Target Use Case
Standard Support	Business Hours (Mon–Fri, 9:00 AM – 6:00 PM local time EST)	Covers routine maintenance, standard inquiries, deployment releases, and L1–L3 support during	Baseline operations and non-critical workloads.

		normal business days.	
Extended Support (12×5)	Mon–Fri, 12 hours a day	Extended-hours monitoring and L2/L3 support, performance monitoring and security updates.	Workloads with extended business hours.
Extended / Global Support	24/5 or 24/7 Coverage	Continuous monitoring and on-call engineering support for critical production issues, high-volume transaction processing, and global users.	Mission-critical platform operations with zero-tolerance for extended downtime.

17.3 Service Level Agreements (SLAs) & Incident Response

Incidents are classified by severity to govern response and resolution timeframes:

The 24/7 response commitments below apply under the Extended / Global plan. Under the Standard plan, response and resolution clocks run only during Standard business hours.

Severity Level	Definition	Response Time	Resolution / Workaround Target
Sev 1 (Critical)	Complete system outage, major security breach, core investment transaction failure affecting all users, or data corruption.	**< 15 minutes** (24/7 on-call)	< 4 hours to restore service or provide a workaround (immediate continuous engineering focus); root-cause analysis within 5 business days
Sev 2 (High)	Partial service degradation, key business module (e.g., Underwriting or Pool Management) impaired, or primary integration failure with workaround unavailable.	**< 1 hour**	**< 8 hours**
Sev 3 (Medium)	Non-critical feature malfunction, minor reporting discrepancy, or isolated user issue with an available workaround.	**< 4 business hours**	**< 2 business days** (or next scheduled patch release)
Sev 4 (Low)	General questions, cosmetic UI feedback, documentation inquiries, or enhancement requests.	**< 1 business day**	Scheduled in upcoming release cycle

17.4 Cloud Operations & Maintenance (CloudOps / DevOps)

Leveraging AWS managed services ensures robust infrastructure management, complemented by our continuous engineering practices:

- **Proactive Monitoring & Alerting:** Automated health checks and performance thresholds configured via Amazon CloudWatch, triggering instant notifications (Amazon SNS) to the on-call engineering team upon anomaly detection.
- **Patch & Vulnerability Management:** Regular patching container images, Lambda runtimes, libraries, and dependencies (automated scanning via CI/CD pipelines and Amazon Inspector).
- **Backup & Disaster Recovery (DR):**
    - Automated daily snapshots and point-in-time recovery (PITR) configured for Amazon Aurora PostgreSQL.
    - Encrypted object versioning and lifecycle policies for documents stored in Amazon S3.
    - Recovery Point Objective (RPO) and Recovery Time Objective (RTO) aligned with agreed business criticality.
- **CI/CD & Release Management:** Controlled deployment pipelines using Infrastructure as Code (AWS CDK / Terraform) to ensure repeatable, audited, and seamless promotion of updates from Dev → Test/UAT → Production.

17.5 Continuous Improvement & Operational Governance

- **Monthly Service Reviews (MSR):** Regular reporting meetings with Seaport Global stakeholders to review system uptime, incident trends, SLA performance, and open support tickets.
- **Capacity & Performance Tuning:** Periodic reviews of AWS resource utilization (compute, database IOPS, API Gateway throttling) to optimize performance and control cloud infrastructure costs.
- **Runbooks & Documentation:** Delivery of comprehensive operational runbooks, architecture maps, and troubleshooting guides during the production handover phase.

18. Commercial Proposal Framework

Commercial pricing should be finalised after discovery and scope confirmation. The following structure can be used for the commercial submission.

18.1. High-Level Cost & Timeline Summary

Module Name	Scope Overview	Estimated Hours	Cost Range (USD)
Module 1: Ingest	Data Onboarding, File Parsing, OCR & Pipeline	80 – 120 hrs	$8,000 – $14,400
Module 2: Underwrite	Credit Scoring, Risk Engine, Financial Modeling	110 – 160 hrs	$11,000 – $19,200
Module 3: Pool	Asset Aggregation, Tranching & Concentration Rules	90 – 130 hrs	$9,000 – $15,600
Module 4: Fund	Capital Calls, Investor Allocations & Disbursement	100 – 150 hrs	$10,000 – $18,000
Module 5: Service	Loan/Asset Servicing, Repayments & Ledger Engine	120 – 180 hrs	$12,000 – $21,600
Module 6: Report	BI Dashboards, Investor Portals & Compliance Exports	80 – 110 hrs	$8,000 – $13,200
Total Project Estimate	End-to-End Client Investment & Brokerage Management Platform	580 – 850 hrs	$58,000 – $102,000

18.2. Detailed Module Breakdown

Module 1: Ingest (Data Onboarding & Pipeline)

Focuses on capturing, parsing, and validating structured and unstructured borrower/asset data from multiple sources.

- Automated PDF/Excel document ingestion with OCR text extraction
- API integrations for automated third-party data pulling (credit bureaus, bank statements)
- Data cleansing, normalization, and automated validation rules
- Secure staging database with automated error flagging and exception queues

Estimated Effort: 80 – 120 hours **Estimated Cost:** $8,000 – $14,400

Module 2: Underwrite (Credit Scoring & Risk Engine)

Implements automated risk evaluation algorithms, financial spreading, and decision-support logic for asset qualification.

- Configurable credit scoring models & financial ratio calculation engine
- Risk rating automation and debt service coverage ratio (DSCR) calculations
- Underwriter override approval workflows with complete audit logging
- Integration with document generation for term sheets and credit memos

Estimated Effort: 110 – 160 hours **Estimated Cost:** $11,000 – $19,200

Module 3: Pool (Asset Aggregation & Tranching)

Manages the bundling of underwritten assets into portfolio pools optimized for funding or securitization.

- Dynamic asset grouping based on yield, maturity, geography, and risk rating
- Concentration limit monitoring and automated rule checking
- Multi-tranche structuring (Senior, Mezzanine, Equity tranches)
- Pool-level cash flow modeling and stress testing simulation

Estimated Effort: 90 – 130 hours **Estimated Cost:** $9,000 – $15,600

Module 4: Fund (Capital Allocation & Disbursement)

Orchestrates investor capital subscriptions, capital calls, funding allocations, and automated disbursements.

- Investor ledger management and commitment tracking
- Automated capital call generation and pro-rata allocation logic
- Disbursement engine with bank wire format generation (ACH / Fedwire / SWIFT)
- Real-time liquidity and available dry-powder dashboards

Estimated Effort: 100 – 150 hours **Estimated Cost:** $10,000 – $18,000

Module 5: Service (Asset Servicing & Sub-ledger)

Tracks post-funding operational lifecycle, repayment schedules, interest calculations, and default management.

- Double-entry sub-ledger tracking principal, interest, fees, and escrows
- Amortization schedule generation with variable rate support
- Payment matching, automated delinquency tracking, and late fee processing
- Restructuring and workout scenario modeling tools

Estimated Effort: 120 – 180 hours **Estimated Cost:** $12,000 – $21,600

Module 6: Report (BI Dashboards & Investor Portals)

Provides real-time visibility, automated investor statement generation, and regulatory compliance exports.

- Interactive executive BI dashboards (Portfolio Yield, LTV, Non-performing loans)
- Self-service investor portal for downloading statements and tax forms (K-1s)
- Automated regulatory compliance exports (e.g., SEC, SOC-1 audit trail, tax summary)
- Custom ad-hoc report builder with export to Excel/PDF formats

Estimated Effort: 80 – 110 hours **Estimated Cost:** $8,000 – $13,200

18.3. Implementation Roadmap & Milestones

Pricing is built by module and released commercially in three stages. The delivery phases and gates in Section 12 are governance milestones and do not change the price.

Commercial Release	Target Modules / Milestones	Duration	Estimated Cost
Release 1 (MVP)	Ingest, Underwrite & Basic Reporting	Months 1 – 2	$27,000 – $46,800
Release 2 (Core)	Pool & Fund Modules Integration	Months 3 – 4	$19,000 – $33,600
Release 3 (Scale)	Service Module & Advanced Analytics	Months 5 – 6	$12,000 – $21,600

18.4. Rate Card & Implementation Assumptions

• **Blended Hourly Rate:** Estimates are calculated based on a blended rate of $100 – $120/hour for senior engineering, architecture, and project management.

• **Architecture & DevOps:** Includes initial cloud environment provisioning (AWS Infra via IaC), database modeling, and CI/CD setup.

• **Testing & QA:** All module estimates include ~20% allocation for QA automation, unit testing, and user acceptance testing (UAT).

• **Out of Scope:** Third-party API license fees, AWS infrastructure usage fees, and ongoing maintenance retainers are excluded.

Also out of scope: Brokerage execution capabilities (Section 3.2) unless confirmed, and additional data-migration sources.

Taxes: Fees exclude VAT, sales tax and any withholding taxes.

Validity: This proposal is valid for 60 days from the date of issue.

AWS consumption costs should be estimated using expected users, TPS, data volumes, storage growth, retention, availability requirements and integration traffic. A detailed AWS pricing exercise should be performed after the workload model is agreed.

The AWS Cost Sheet referred to in Section 18.5 is not part of this version. It will be issued once Appendix B inputs are received.

18.5. Warranty & AMC Pricing Plan

I. Warranty Period

Essarsystems provides a comprehensive warranty to ensure platform stability, defect resolution, and seamless post-launch transition.

- **Warranty Duration: 365 Days (12 Months)** included post-production Go-Live and Sign-Off.
- **Scope of Coverage:**

- **Defect Rectification:** Correction of software bugs, functional defects, or unexpected behaviors in custom-built code (React frontend, Python, Spring Boot / AWS Lambda services).
- **SLA Compliance:** Priority-based resolution of defects raised on production issues during the warranty period per Section 17.3, under the coverage plan selected. Warranty covers defects only; operational support (L1–L3 monitoring, CloudOps) is provided under the support option selected.
- **Performance Monitoring:** Continuous monitoring via AWS CloudWatch to ensure optimal system performance under initial production loads.

- **Exclusions:**
  - New feature requests or scope additions outside agreed-upon functional requirements.
  - Third-party service/API failures beyond Essarsystems' control.
  - AWS cloud infrastructure usage costs billed directly by AWS.

II. Annual Maintenance Contract (AMC) / Post-Warranty Support Models

Following the warranty period, Seaport Global can select from flexible support models tailored to business needs:

Option A: Percentage-Based AMC (Standard Industry Model)

- **Pricing: 18% – 20%** of total software development / implementation cost annually.
- **Billing Frequency:** Billed **Quarterly in Advance**.
- **Inclusions:**
  - Comprehensive L2 & L3 support coverage.
  - Routine bug fixes, minor security patches, and minor enhancements.
  - AWS CloudOps, dependency/library updates, and CI/CD maintenance.
  - Monthly service review and SLA compliance reporting.

Option B: Tiered Monthly Support Retainer

Support Tier	Scope & Coverage	Monthly Fee (USD)	Included Monthly Hours	Overage Rate / Hr
Basic Support	L2 Support, Business Hours Coverage (9x5), Routine Maintenance & Bug Fixes (Standard Support plan)	$1,600 / month	Up to 40 hours	$40 / hour
Standard Support	L2/L3 Support, Extended Hours Coverage (12x5), Performance	$2,800 / month	Up to 80 hours	$35 / hour

	Monitoring, Security Updates (Extended Support 12x5 plan)			
Enterprise Support	Dedicated L2/L3 Engineering Support, 24/7 Severity-1 Incident Response, Monthly SLA Reviews, CloudOps (Extended / Global Support plan)	$4,500 / month	Up to 150 hours	$30 / hour

Option C: Time & Materials (T&M) On-Demand Support

For ad-hoc support or custom feature additions outside regular maintenance:

Role	Standard Hourly Rate
Solution / Cloud Architect	$110 / hour
Senior Backend / AWS Engineer	$90 / hour
Support / DevOps Engineer (L2/L3)	$80 / hour
React Frontend Engineer	$75 / hour

III. Commercial Terms & AWS Infrastructure Costs

1. **Third-Party & AWS Cloud Costs:** Billed directly to Seaport Global via their AWS account. AWS infrastructure estimates are provided separately in the AWS Cost Sheet.
2. **Invoicing & Payment Terms:** AMC payments are billed quarterly in advance with net 30-day payment terms.
3. **Annual Revision:** AMC pricing is subject to an annual indexation adjustment (CPI max 3–5%) at contract renewal.

18.6 Onsite Travel & Out-of-Pocket Expenses Clause

I. Scope of Fixed Commercials

Unless explicitly stated as an inclusive fixed-fee deliverable, all professional fees quoted in this proposal cover remote offshore/nearshore service delivery, engineering, cloud provisioning, and project management.

II. Client-Requested Onsite Visits

Any onsite presence requested by **Seaport Global** including but not limited to discovery workshops, architecture reviews, business stakeholder alignment, user acceptance testing (UAT) support, or go-live hypercare shall be subject to prior mutual written approval.

III. Reimbursable Expense Terms

All travel, lodging, and incidental expenses incurred as a direct result of approved onsite travel will be reimbursed by **Seaport Global** at actual cost, subject to the following guidelines:

- **Travel & Airfare:** Economy class for domestic flights; Premium Economy or Business Class for international flights exceeding six (6) hours of continuous flight time.
- **Accommodation:** Standard single occupancy room in a business-tier hotel (3 - 4 star equivalent or near Seaport Global office locations).
- **Ground Transportation:** Business mileage (at standard local tax authority rates), airport transfers, train fare, or ride-share services (e.g., Uber/Taxi) directly related to the engagement.
- **Meals & Incidentals (Per Diem):** Actual reasonable meal expenses or an agreed daily per diem rate per resource per day.

IV. Approval & Invoicing Procedure

- **Pre-Approval:** Essarsystems will submit an estimated travel itinerary and cost budget to Seaport Global's designated Project Manager for written approval prior to booking.
- **Receipts & Invoicing:** Reimbursable expenses will be invoiced monthly in arrears accompanied by itemized receipts and supporting documentation.
- **Payment Terms:** Expense invoices shall be payable within **30 calendar days** from the invoice date.

18.7 Insurance Coverage Clause

I. Vendor Insurance Maintenance

Essarsystems agrees to maintain in full force and effect, at its own sole cost and expense throughout the duration of this agreement (including any warranty and subsequent maintenance periods), comprehensive insurance policies issued by reputable carriers licensed to operate in the relevant jurisdictions. Limits are to be verified against current policy schedules before issue.

II. Required Coverage Levels

Essarsystems maintains the following policy limits, specifically structured for institutional capital management, brokerage technology, and cloud-native software implementations:

- **Professional Indemnity / Technology Errors & Omissions (E&O) Insurance:**
    - **Coverage Limit: £5,000,000** (or **$5,000,000 USD**) aggregate / per claim. (USD equivalent at the prevailing exchange rate).
    - **Scope:** Protects against financial losses arising from software defects, design errors, implementation failure, breach of professional duty, or performance failures.
- **Cyber Security & Data Breach Liability Insurance:**
    - **Coverage Limit: £5,000,000** (or **$5,000,000 USD**) aggregate. (USD equivalent at the prevailing exchange rate).

- **Scope:** Covers liabilities resulting from data security breaches, unauthorized system access, cyber extortion, data recovery costs, and notification expenses.

- **Public & General Liability Insurance:**
  - **Coverage Limit: £2,000,000** per occurrence / aggregate.
  - **Scope:** Protects against third-party bodily injury or property damage claims arising during project execution or onsite activities.

- **Employer's Liability Insurance:**
  - **Coverage Limit: £5,000,000** aggregate (the UK statutory minimum).
  - **Scope:** Covers legal liability for injury, illness, or disease sustained by employees during the engagement.

III. Proof of Insurance & Certificates

Upon request or execution of the formal Master Services Agreement (MSA), Essarsystems will provide **Seaport Global** with valid Certificates of Insurance (COI) demonstrating compliance with the specified coverage limits and policy terms.

IV. Limitation of Liability Alignment

The total financial aggregate liability of Essarsystems under the formal agreement shall be aligned with and capped at the available limits of the applicable insurance coverage described above, excluding willful misconduct, fraud, or intentional breach of confidentiality obligations. Any further carve-outs are to be set out in the MSA.

18.8 Other Commercial Terms

The following terms are placeholders to be completed with Essarsystems commercial and legal teams and to be defined in the MSA.

- Intellectual property: Seaport Global owns bespoke deliverables, Essarsystems retains pre-existing IP and accelerators
- Change control: changes to scope, timeline or price require written approval following an impact assessment.
- Termination and exit assistance: [notice period and transition services to be agreed], including handover of source code, IaC and documentation.
- Data protection: a Data Processing Agreement will be executed before any personal data is processed.
- Governing law and dispute resolution: [to be agreed].

19. Acceptance Criteria

- All agreed MVP functional requirements are implemented and traceable to approved acceptance criteria.
- Critical and high-severity defects are resolved or formally accepted before production.
- Security testing meets agreed acceptance thresholds.
- Performance testing demonstrates agreed response-time and transaction-volume targets.
- Production monitoring and alerting are operational.
- Backup and recovery procedures are documented and tested to the agreed level.
- Operational documentation and knowledge transfer are completed.
- Business stakeholders complete UAT and provide formal acceptance.
- Migrated data is reconciled to the agreed tolerance and signed off by the business.
- The DR test meets the agreed RPO and RTO.
- Accessibility testing meets the agreed standard (WCAG 2.1 AA).
- The independent penetration test has no open critical or high findings.

20. Conclusion

The proposed Client Investment & Brokerage Management Platform provides a modern foundation for digitising client investment operations in Seaport Global. The combination of React, AWS managed services, modular business capabilities, API-led integration and event-driven processing provides a scalable and secure architecture that can evolve with future investment products and services.

The recommended next step is a focused discovery and architecture phase to validate business processes, investment/product rules, transaction volumes, integration landscape, regulatory requirements, data migration needs and target service levels. The outputs of this phase will establish the detailed MVP scope, delivery roadmap, resource plan and final commercial estimate. Gate 2 confirms the final price for the build.

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Appendix A – Proposed AWS Component View

Capability	AWS Services / Technology
User Experience	ReactJS, S3, CloudFront
API Management	API Gateway, WAF
Business Services	ECS/Fargate with Spring Boot (core services); AWS Lambda (event handlers and adapters)
Workflow	AWS Step Functions
Document intelligence	Amazon Textract
Messaging	SQS, SNS, SES
Events	EventBridge
Transactional Data	Aurora PostgreSQL, RDS Proxy
Documents / Archive	S3 (versioning, Object Lock, replication)
Identity	Cognito and/or Enterprise Identity Provider
Security	IAM, KMS, Secrets Manager, WAF, Shield, CloudTrail, GuardDuty, Security Hub, Config, Macie, Inspector
Monitoring	CloudWatch, X-Ray / OpenTelemetry
Networking	VPC, private subnets, Security Groups, VPC endpoints
Landing zone	Organizations, Control Tower, IAM Identity Center
Backup and DR	AWS Backup, cross-Region replication
CI/CD	CodePipeline, CodeBuild and/or GitHub Actions
Infrastructure	CDK / Terraform
Analytics	QuickSight, Athena, Glue (Redshift if volumes warrant)

Appendix B – Information Required for Detailed Estimate

- Expected number of clients and internal users
- Expected daily/monthly transactions
- Peak TPS and concurrency
- Number and type of investment products
- Number of funds/pools
- Expected document volume and annual storage growth
- Required data retention period
- Existing systems and third-party integrations
- Current application/API landscape
- Legacy data migration requirements
- Availability, RPO and RTO requirements
- Regulatory and reporting requirements
- Target countries/regions and data residency requirements

- Expected support hours and SLA requirements
- Target jurisdiction(s) and applicable regulators
- Whether brokerage execution capabilities are in scope
- Existing AWS organisation, accounts and landing zone, if any
- Enterprise identity provider and single sign-on requirements
- Number of legacy sources to migrate and data volumes
- Service levels required by downstream systems